

Name: _____ () CT Group: 22S0 _____

RAFFLES INSTITUTION

2022 YEAR 6 TERM 2 TIMED PRACTICE

22 March 2022

H2 PHYSICS

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Part 2

INSTRUCTIONS TO CANDIDATES

Write your name, index number and CT Group on the cover page.

Write your answers to Part 2 in the spaces provided on the question paper.

For Examiner's Use		
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This document consists of 8 printed pages.

Part 2 (32 marks)

- 6 (a) A uniform aluminium strip is clamped at one end. A metal block of mass M is attached to the free end of the strip at a distance L from the clamp. The mass causes the free end of the strip to be depressed by a distance d . Fig. 6.1 shows the strip and the mass at the equilibrium position.

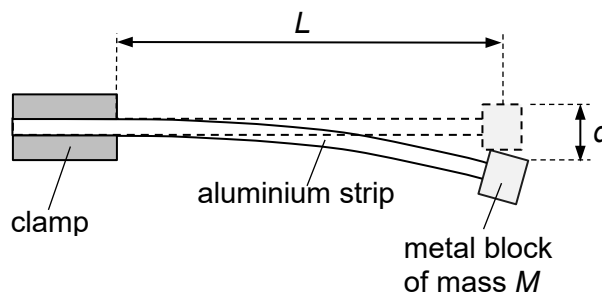


Fig. 6.1

The free end of the strip is then given a small vertical displacement and released.

The variation with vertical displacement x from the equilibrium position of the acceleration a of the free end of the strip is given by the expression

$$a = -\frac{kE}{L^3 M} x$$

where k is a constant for the strip and E is a constant for the material of the strip known as Young's modulus.

- (i) Explain why the expression in (a) leads to the conclusion that the strip is performing simple harmonic motion.

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.....

..... [2]

- (ii) The Young's modulus E for aluminium is $7.0 \times 10^{10} \text{ N m}^{-2}$. Given that M is 0.12 kg and L is 0.60 m, the variation with time t of the depression d of the free end of the strip as it oscillates, is given by

$$d = 3.0 + 1.2 \cos(34.9t)$$

where d is in cm and t is in seconds.

1. State the initial depression of the strip before it was made to oscillate.

initial depression = cm [1]

2. State the amplitude of the oscillations.

amplitude = cm [1]

3. Calculate the frequency of the oscillations.

frequency = Hz [1]

4. Determine the value (without the unit) of k for this strip.

value of k = [2]

- (b) Fig. 6.2 shows the variation with frequency f of the amplitude x_0 of the forced oscillations of a machine.

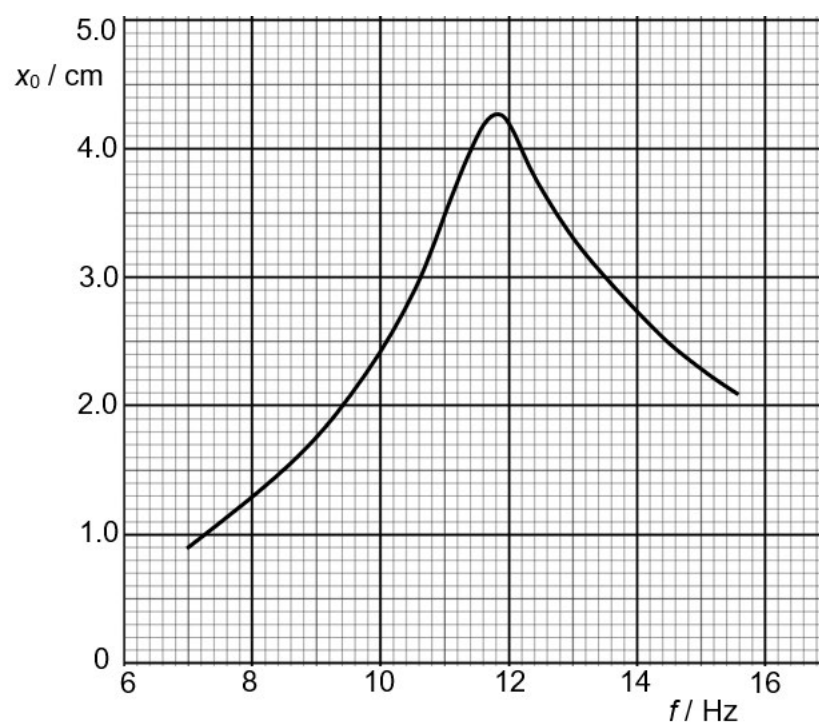


Fig. 6.2

- (i) State

1. what is meant by a *forced oscillation*,

.....

 [1]

2. the name of the effect shown in Fig. 6.2.

..... [1]

- (ii) At any value of the frequency, the oscillations of the machine are simple harmonic.

Calculate, for the machine vibrating at maximum amplitude,

1. the maximum speed,

maximum speed = m s^{-1} [2]

2. the maximum acceleration,

maximum acceleration = m s^{-2} [2]

3. the time interval between a maximum speed and the subsequent maximum acceleration.

time interval = s [1]

- (iii) The heavy-duty springs on which the machine is mounted on are replaced by springs with a smaller force constant.

On Fig. 6.2, sketch the new graph for the variation with frequency f of the amplitude x_0 of the forced oscillations of the machine.

[2]

- 7 (a) State the *principle of superposition*.

.....

.....

..... [2]

- (b) Coherent light of wavelength 589 nm is incident normally on a double slit arrangement, as shown in Fig 7.1.

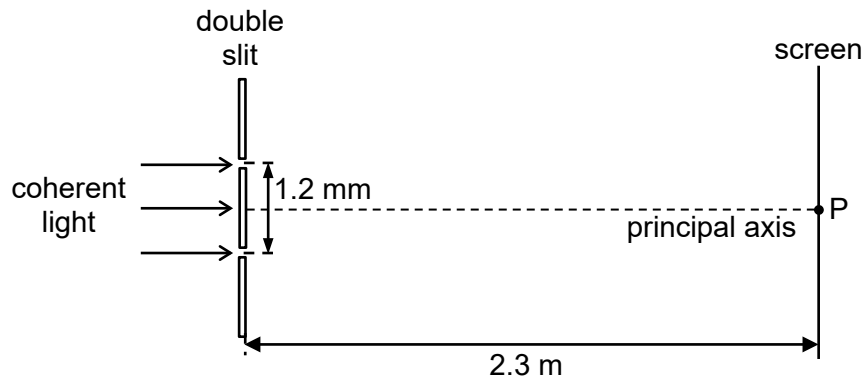


Fig. 7.1 (not to scale)

The separation of the slits is 1.2 mm.

Interference fringes are observed on a screen placed parallel to the plane of the double slit and 2.3 m from it. The centre of the interference pattern is at point P.

- (i) Explain how interference fringes are formed along the screen.

.....

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.....

..... [2]

- (ii) It is necessary for the light from the two slits to be coherent to form a well-defined interference pattern on the screen.

1. State the meaning of the term *coherent*.

..... [1]

2. State another necessary condition for the formation of a well-defined interference pattern.

..... [1]

- (iii) Determine the separation of the bright fringes of the double slit interference pattern.

separation = m [2]

- (iv) The width of each slit in Fig. 7.1 is 0.30 mm.

The double slit interference pattern on the screen is observed to have some missing bright fringes.

1. Show that the two lowest orders of the bright fringes that are missing are the fourth and the eighth orders.

[2]

2. On Fig. 7.2, sketch a graph to show the variation with displacement from point P of the intensity of the light on the screen up to the eighth order bright fringes.

With the help of your answer in (b)(iii), include appropriate values on the horizontal axis.

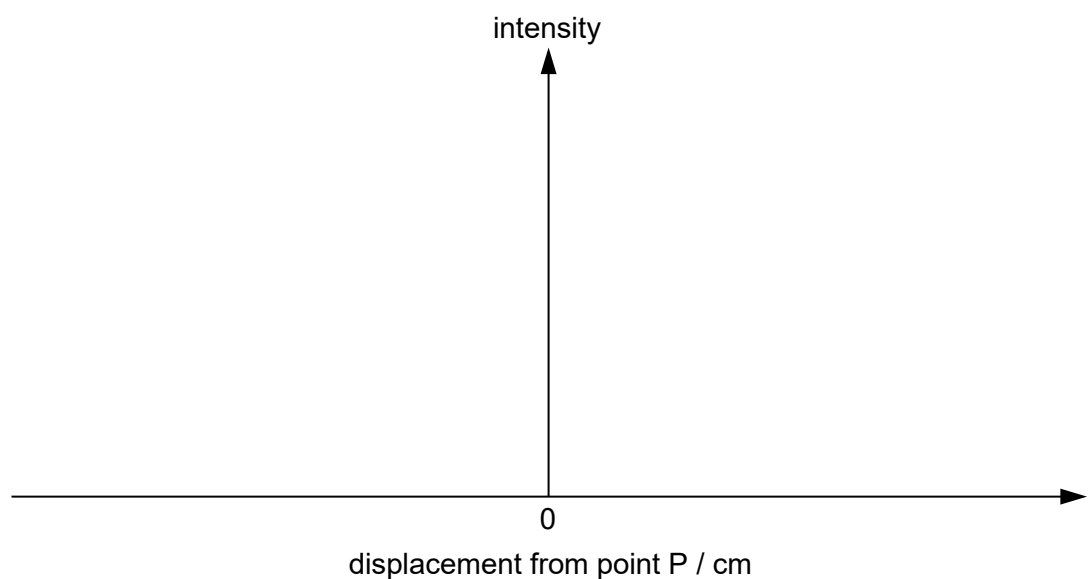


Fig. 7.2

[2]

- (v) The double slit in Fig 7.1 is replaced with a diffraction grating that has 500 lines per millimetre.
1. Determine the number of orders of diffracted light that can be observed on each side of P.

number of orders = [2]

2. Explain why the interference pattern produced by light passing through a diffraction grating is brighter than that produced by light from the same source passing through a double slit.

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.....
..... [2]

END OF PART 2